Pseudo-code of 4-Bit Counter Simulator

1. Start the Program

START

2. Initialize Variables

Set counter value = 0

Set timer = NULL

3. Initialize the Display

Convert the counter value into a 4-bit binary number.

Display the binary value on the webpage.

Display the decimal value on the webpage.

Set all four LEDs according to the binary bits.

4. Update Display Function

FUNCTION updateDisplay()

Convert the current counter value into binary.

Make sure the binary number contains exactly four bits.

Display the binary number.

Display the decimal counter value.

IF the first binary bit is 1, turn ON LED 1.
ELSE turn OFF LED 1.

IF the second binary bit is 1, turn ON LED 2.
ELSE turn OFF LED 2.

IF the third binary bit is 1, turn ON LED 3.
ELSE turn OFF LED 3.

IF the fourth binary bit is 1, turn ON LED 4.
ELSE turn OFF LED 4.

END FUNCTION

5. Up Counter Function

FUNCTION upCounter()

Increase the counter value by 1.

IF the counter reaches 16, set it to 0.

Call updateDisplay() to update the webpage.

END FUNCTION

6. Down Counter Function

FUNCTION downCounter()

Decrease the counter value by 1.

IF the counter goes below 0, set it to 15.

Call updateDisplay() to update the webpage.

END FUNCTION

7. Reset Counter Function

FUNCTION resetCounter()

Set the counter value to 0.

Call updateDisplay() to show the initial values.

END FUNCTION

8. Automatic Counting Function

FUNCTION startAuto(direction)

Stop any previously running automatic timer.

Start a timer that repeats after every 1 second.

IF direction is UP:
Call upCounter().

ELSE:
Call downCounter().

END FUNCTION

9. Stop Automatic Counting Function

FUNCTION stopAuto()

Clear the automatic timer.

Set timer = NULL.

Stop automatic counting.

END FUNCTION

10. Display Initial Values

Call updateDisplay() when the program starts.

Display binary value 0000.

Display decimal value 0.

Turn OFF all four LEDs.

11. End

END PROGRAM